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GaLaPaGoS: A Design Pattern for Sustainability of ICT Interactive Software and Services

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Abstract: ICT is an enabler for sustainable development and contributes to sustainability problems arising from different human activities aided by ICT interactive software and services. Consequently, these ICT-aided activities contribute to a vast amount of electronics and paper package waste at different stages of production, usage, and disposal. Some of these wastes can be attributed to a lack of understanding from ICT designers on how to design ICT interactive software and services for a green user experience. This paper explores design patterns as a tool to capture and incorporate sustainability concerns into the design of ICT interactive software and services for green user experience, where users become more conscious of the sustainability impacts of their actions. The Design science research method was applied to create and validate three sustainability design patterns. The results show that the proposed design patterns can guide ICT designers to engage and influence user behavior toward sustainability awareness.

Keywords: ICT, Green UX, Sustainability awareness, persuasive technology, e-waste, eco-feedback, repair, reuse

1 Introduction

The concern for sustaining our world is increasing worldwide [1] and the importance of sustainability is highlighted by the United Nations Sustainable Development Goals (SDGs) [2]. Human activities and actions are central to some of the sustainability problems today with the continuous usage of Information and Communication Technology (ICT) and software systems, which have led to the disposal of different electronic waste (e-waste) [3] and paper waste. For example, in Europe, 16 tons of materials (glass, plastic, wood, papers, etc.) are used per person yearly, out of which 6 tons become waste. From this total of 2.5 billion tons, only a limited (albeit increasing) share (36%) was recycled, with the rest landfilled or burned, of which some 600 million tons could be recycled or reused [4]. According to E-waste Monitor, 50 million tons of e-waste are generated annually [5] of which a sizeable proportion is related to ICT. Fig. 1 illustrates that e-waste is mostly generated in developed countries and

transferred to developing countries in Africa and Asia, where e-waste management is still a problem [6].

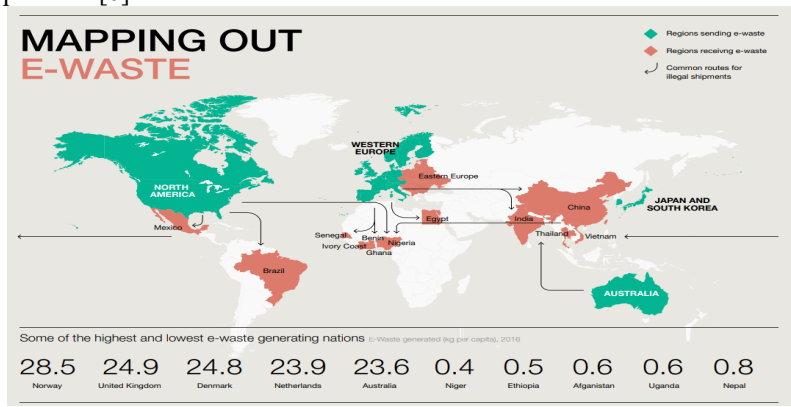


Fig. 1. The global E-waste statistics on e-waste generated by nations adapted from [5]

An investigation for Sunday's Dateline [7] found that varieties of old electronic equipment were discarded in what was once a picturesque wetland of Ghana, West Africa, without proper waste management. Research outcomes, as presented in [8][6][9][10], show increased concerns and efforts toward proper ICT e-waste management in developing and developed countries [11] through extended producer responsibility (EPR) and efficient recycling facilities but less research on user behavioral change towards waste reduction (e-waste and paper waste) aided by user interaction with ICT interactive software and services. The US Environmental Protection Agency reported that humans are at the center of waste generation [12] as most of their activities impact sustainability (environmental, social, individual, and economic). As such, ICT interactive software and services play a significant role in waste reduction (e-waste and paper waste). Therefore, there is a need to investigate how to design sustainably and develop ICT interactive software and services to support user behavior towards reducing waste and emissions and promote a mindset change towards repair, reuse, and recycling options among users.

The following scenario is an elaborate typical problem of user experience with current ICT interactive software and services:
"Many mobile phone devices and computers we buy are packaged in nice and very attractive boxes. What do we do with boxes after buying and using the devices? What do we do with the device at the end of life (EOL)? The user experience (UX) analysis shows that the boxes may stay in users' drawers for years, or users may trash them. Also, the device at the EOL might be disposed because of bad parts. What if the manufacturer has a Web service that provides users with information about alternative ways of using the device parts at EOL and the benefits of reusing the box used in packaging the device? UX analysis again shows that users may not access it and thus never read this information. Then the question is, how do we design the accessibility and the attractiveness of the Web service for green user experience (green UX)?"

From the context of this research, green UX refers to the effectiveness of user interaction with interactive ICT software and services in creating user awareness about their environmental and societal impacts on sustainability. It entails how well ICT interactive software and services support change in user behavior toward reducing resource wastage and encouraging repair, reuse, and recycling. From the above scenario, green UX will be that whenever a user opens the box and starts the device, the web service is integrated into the device setup operations. As part of the setup service, the device tells the user how to easily reuse the package box, the device at

As such, these different studies show the need to incorporate green UX in the design of interactive ICT software and services because green UX will aid better changes in user habits, which over time can support user behavior towards sustainability (creating user consciousness about their impacts on sustainability, decrease in resource wastage such electronic and paper waste) with good eco-feedback on their action.

3 Research Design

This research is grounded in the overall design science research methodology (DSRM) [33] that provides the process of iteratively creating and testing artifacts, which were applied in this research to solve problems of green UX in ICT interactive software and services. Fig. 2 provides an overview of how the three design science research cycles (Relevance Cycle, Design Cycle, and Rigor Cycle) from [33] were applied in the study.

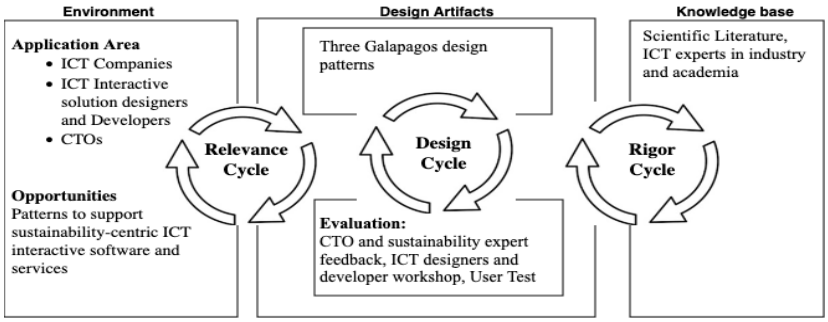


Fig. 2. Design science research cycles adapted from Murugesan and Gangadharan [33]

- **Relevance Cycle**: The relevance cycle initiates design science research within the context of finding solutions to the challenge of persuading and influencing positive user behavior towards sustainability (creating user consciousness about their impacts on sustainability and decreasing electronic and paper waste) because of the usage of different ICT interactive software and services. The relevance cycle (contextual environment) is connected to the design cycle (construction of artifact) to ensure a better evaluation of the created artifacts and determine whether additional iterations of the relevance cycle are needed in this design science research.
- **Rigor Cycle**: Design science research requires knowledge based on scientific theories that provide a foundation for the research work. Within this research, the rigor cycle covers studying past knowledge on sustainability problems of ICT interactive software and services with artifacts (tools, processes, guidelines) that already exist to tackle the problem of persuading and influencing positively user behavior towards sustainability through user sustainability awareness.
- **Design Cycle**: This is the heart of design science research because this cycle of research activities iterates more rapidly between the construction of an artifact, its evaluation, and subsequent feedback to refine the design artifacts based on the feedback. The design cycle in this research covers activities such as workshops

and evaluation of all the created artifacts with software development practitioners in the industry.

4 Research Execution and Results

This section covers the creation and validation of the design patterns (GaLaPaGoS design pattern) by software development practitioners and end-users testing prototypes created based on GaLaPaGoS design pattern to support green UX.

4.1 Relevance Cycle 1: Problem Identification

Research results from literature studies in the background section highlight different problems for designing ICT interactive software and services to support positive user behavior towards reducing waste (electronic and paper waste) and user consciousness about their impacts on sustainability. Therefore, a full-day participatory pattern workshop [34] with software development practitioners was organized. Table 1 provides the background details (Job title and years of experience) of the software development practitioners (workshop participants) from different companies in the industry.

Table 1. Background of workshop participants from different companies

No	Job Title	Years of Experience
1	CTO	13
2	UX Lord	11
3	Programmer	8
4	UI Designer	6
5	ICT Engineer	5
6	DevOps Manager	5
7	Software Developer	4
8	IT Manager	4
9	Business Analyst	4

The researchers (authors) acted as facilitators and participants in the workshop. The first phase of the workshop focused on identifying design challenges related to green user experience (UX) in ICT interactive software and services. The workshop participants highlighted several key issues. Firstly, they noted a lack of practical guidelines and design tools suitable for integrating green UX into the design process. Existing sustainability guidelines were considered impractical for business environments because it is time-consuming and not economically feasible. Moreover, the workshop participants emphasized the need for practical examples demonstrating how sustainability can be effectively translated into design elements of interactive software and services to enhance user consciousness of their sustainability impacts. Additionally, they highlighted a gap in incorporating sustainability into software development and design practitioners education and training. Regarding resource waste from user interactions with ICT interactive software and services, major concerns included excessive printing leading to paper and energy waste in companies, electronic waste

from discarded devices (computers, smartphones, tablets, etc) and packaging waste from e-commerce activities. Addressing these challenges requires the development of design tools that support designers in creating green UX solutions while considering business constraints and sustainability complexities.
Next Action: Develop a tool to support and educate ICT interactive software and services designers to design ICT interactive software and services for green UX.

4.2 Rigor and Design Cycle 2: Development of Artifacts (Patterns)

The second phase of the workshop involves co-designing solutions for all identified green UX problems in interactive software and services. Researchers and workshop participants explored persuasive design tools to support green UX and concluded that design patterns serve as a structured template to support designers of ICT interactive software and services for green UX. This led the researchers to introduce participatory pattern workshop [34] guidelines to facilitate different design narratives with the workshop participants on ICT interactive software and services for green UX. This resulted in the creation of three GaLaPaGoS design patterns (Improved packaging, green printing, and electronic lifecycle extension). Table 2 outlines the pattern template used in creating the three GaLaPaGoS design patterns by the researchers and workshop participants.

Table 2. Patterns template adapted from participatory pattern workshops resource kit [34]

Pattern content	Description
Name	Naming is important. Think of a short, catchy phrase that captures the essence of your pattern.
Summary	Try to capture the essence of the pattern in 2-3 sentences. Focus on function - what it does, not how it is built.
Illustration	Metaphoric or inspirational image, which captures the spirit of the pattern.
Problem	What is the problem that this pattern addresses? What does it try to achieve?
Context	When and where is this pattern most relevant?
Solution	Describe the core of the solution in such a way that it can be directly implemented a million times without doing the same thing twice.
Diagram	A structural or narrative graphic that supports the detailed description of the solution.
Related Patterns	List other patterns related to this one under categories such as component, assisting, conflicting, uses this, etc.

4.2.1 The Electronic Lifecycle Extension Pattern

The Electronic Lifecycle Extension pattern, as detailed in Table 3, focuses on electronic waste, which is a global ecological issue because of the increase in the amount of e-waste from increased usage of different electronic devices [35] to access different ICT interactive software and services for our daily activities.

Table 3. GaLaPaGoS pattern template for electronic lifecycle extension pattern

Pattern Name	Electronic Lifecycle Extension
Context/Intent	This pattern showcases how to repair hardware components (screen, battery, memory, processor) to avoid disposal of usable electronics, and when an electronics device is damaged, usable components are recommended for reuse by the person in need of such components.
Problem	Electronics are disposed of even when components (screens, chips, metals, plastics) are still working or can be repaired.
Forces	• Relatively new electronics such as cell phones are replaced by new ones often within 6 months to 2 years. • Electronic components have a long life expectancy. • Materials are not biodegradable and hard to extract.
Solutions	• Show the life expectancy of components. • If some component in an electronic device is damaged, suggest repair/replacement. • Provide guides to repair your cell phone. • Provide sources where to find parts. • Encourage repair with motivational phrases. • Provide solutions if repair is not effective.
Consequences	• Fewer electronics are disposed off in the environment. • Increased life expectancy of electronics. • Users become more aware of their environmental impact. • Reduced environmental pollution.

The electronic lifecycle extension pattern supports the reuse of electronic components (screen, battery, memory, processor) to reduce the disposal of usable electronics, and if disposed of, reuse recommendations for a person in need of those components should be available.

4.2.2 The Green Printing Pattern

Printing is a function in ICT interactive software and services that serve as a source of paper waste in companies and homes. However, the consequential effects of printing have cross-boundary impacts on the environment: trees used in producing papers, energy consumed in printing and reprinting the same documents, paper wastage and hardware resources (printers) used in printing, the exploitation of forests to obtain pulp for the paper, to the use of inks that pollutes the environment. Billions of tons of paper are used worldwide (Statistic Brain Research Institute, 2016) and the number keeps growing due to the availability and reduced prices of printers and paper in the world.

Table 4. GaLaPaGoS pattern template for green printing pattern

Pattern Name	Green Printing
Context/Intent	This pattern is to reduce/stop printing unless necessary.
Problem	Large sums of documents are printed unnecessarily generating a huge

	waste of resources (paper and ink) and having a negative impact on the environment.
Forces	• The document size is large and is widely used, and almost every person in the organization wants to print. • The document is in colours, 1 face, spaces between lines and margins are wide • Oversized images that take up much space.
Solutions	• Disable by default the print button • If a user insists on printing, display a warning • Show leader board/point reward system • Finally allow print but with print, settings changed.
Consequences	• Users print less • Fewer resources are consumed.

Table 4 provides details of the green printing pattern centered on the design problems associated with printing in different interactive software and services which can be used by designers of green UX to reduce paper waste and make users aware of their sustainability impacts from printing.

4.2.3 Improved Packaging Pattern

This pattern aims to support the design of an online shopping experience where users are given choices of using less paper packaging and shown how to reuse their package boxes for the alternative purpose of reducing paper waste that comes from online shopping.

Table 5. GaLaPaGoS pattern template for improved packaging pattern

Pattern Name	Improved Packaging
Context/Intent	This pattern displays how to reduce unnecessary extra packing when ordering items online.
Problem	Due to the convenience of online shopping, more people are buying items online, which increases the amount of packaging used for shipping those items.
Forces	• Any item bought online has its own packing plus the packaging for shipping. • People tend to buy items separately. • Some materials (plastic) are not biodegradable. • The package is disposed off without recycling or reuse.
Solutions	• When purchasing the item online, show the types of packaging available for shipping, and pre-select less packaging, thus a cheaper option. • Attach instructions to conserve the packaging and reuse them as gift boxes. • Display instructions to encourage recycling. • Encourage the devolution of packages for further use in other shipping.

Consequences	• Use only necessary packages when sending items. • More reused packages within online shopping. • Customers reuse packages for other purposes.

The improved packaging pattern (See Table 5) targets those in charge of the technological and business logic of online shops to help reduce the amount of paper packaging waste from shopping online.
Next Action: Apply the three GaLaPaGoS design patterns to test effectiveness in supporting design decisions of ICT interactive software and services for green UX.

4.3 Relevance Cycle 3: Testing of the artifacts by developers

This section provides results demonstrating the applicability of GaLaPaGoS design patterns to support green UX by designing sustainability-centric interactive ICT software and services. Four software developers with more than five years of experience were recruited from different companies to work together and apply the three GaLaPaGoS design patterns for the redesign of any interactive software or service of their choice.

4.3.1 Green Printing Pattern

A daily activity that is related to ICT in every workplace and household is printing. Printing activities involve different resources, such as hardware, software, and paper. The four developers applied the green printing pattern to reduce unnecessary printing and create user awareness about paper wastage from printing. Google Docs and Microsoft Word documents were used case for the design of a mock-up solution. Fig. 3 shows the design mock-up based on a green printing pattern where users are informed of their printing activities about natural resources (trees) and their CO_2 emissions.

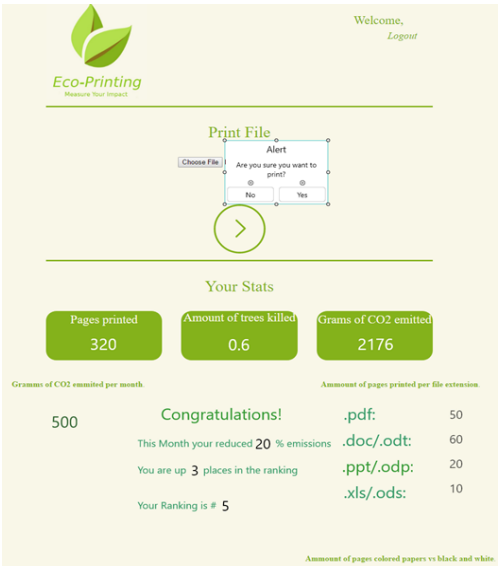


Fig. 3. Design mock-up based on green printing pattern

This interface provides users with a statistical overview of their printing activities and educates them about their impacts on sustainability and the environment through the information in the dashboard (CO2 emission, rank in the company, and relating the printed documents to trees killed in the forest). The four developers indicated that the green print pattern can be applied to these scenarios to promote green UX:

1. The first scenario is when the user wants to reprint an already printed document, a pop-up tip from the interface tells the user there are two or three pages that have changed since the last time a user edited that same document and asks if the user wants to reprint only pages that have changed.
2. The second scenario is that when users want to print with color, a tip from the interface informs users to print only colored pages.
3. Finally, the dashboard gives user statistics about the percentage of their printing and the number of pages, which equals a certain amount of killed trees, to create awareness about their environmental impacts.

Feedback from developers: Add to the green pattern template solution: "alert users of oversized images" and "tips for users to reduce image size".

4.3.2 Electronic Lifecycle Extension Pattern

All our modern-day life requires the use of different electronic devices or electrical equipment such as phones, computers, tablets, and screens. The developers used the electronic lifecycle extension pattern to create a mock-up design (see Fig. 4) for supporting an extension of electronic device lifespan that can be incorporated into electronic device apps like mobile phones, computers, and tablets to create user awareness about how to repair, reuse and replace components of electronic devices.

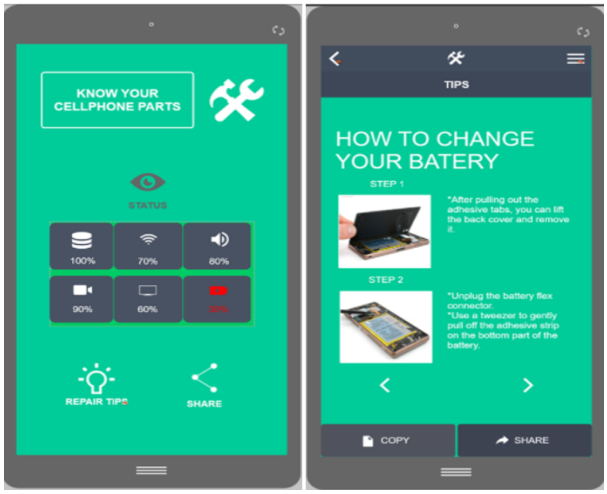


Fig. 4. Design mock-up-based electronic lifecycle extension pattern

According to the four developers, the application of the Electronic Lifecycle Extension pattern in the design and development of ICT solutions can guide end-users on

how to repair, reuse, and fix parts of the electronics instead of disposing the electronic device, which in the long term will help reduce the ecological footprint of ICT waste.
Feedback from developers: Extend this pattern or create another pattern that can aid diverse alternatives for reusing electrical components at the end of the life cycle for other purposes.

4.3.3 Improved Packaging Pattern

One major source of waste comes from items used in packaging products. Items such as cardboard, plastic, and special foams are used to package products that will later become waste.

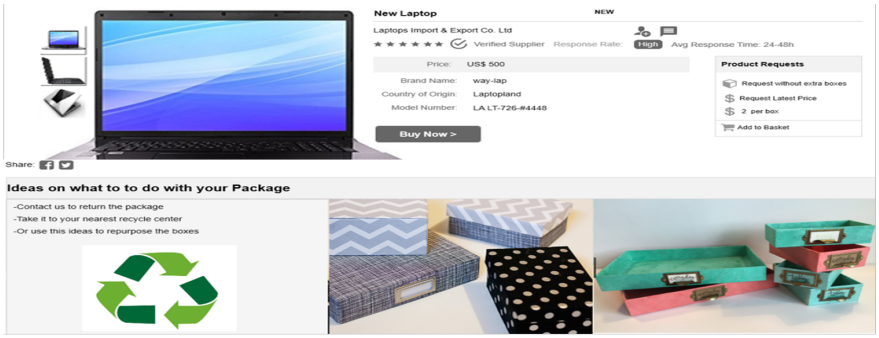


Fig. 5. Design mock-up based on green printing pattern

Though logistics have improved with the use of ICT technology for improved delivery [24], it has increased waste due to poor packaging. The software developers created a mock-up design (see Fig. 5) for online stores like Amazon and eBay using the improved packaging pattern as a guide. The four developers stated that by applying the improved packaging pattern, the design mock-up in Figure 5 is one example of how those in charge of the technological aspect of online retail stores can design purchase experience to inform users on how to reuse package boxes and charge extra for packaging to discourage unnecessary packaging waste.
Feedback from developers: Create another pattern focusing on information display within interactive software and services for green UX that can support user engagement and foster better sustainability awareness.

The design mock-up results from applying the three GaLaPaGoS design patterns show it can serve as a design tool to support designers in adding elements of sustainability to their design decisions for ICT interactive software and services. This can support green UX, where users become aware of their impacts on sustainability.

4.3.4 User awareness campaign

The four developers proposed a prototype idea for user awareness called “Save the Planet Campaign,” which can be incorporated into a mobile phone and tablet setup to improve user’s green UX. The developers combined the electronic lifecycle extension pattern and the improved packaging pattern to design a prototype using Marvelapp

[37], an online prototyping tool to explore the Save the Planet Campaign mimicking a new mobile phone setup.

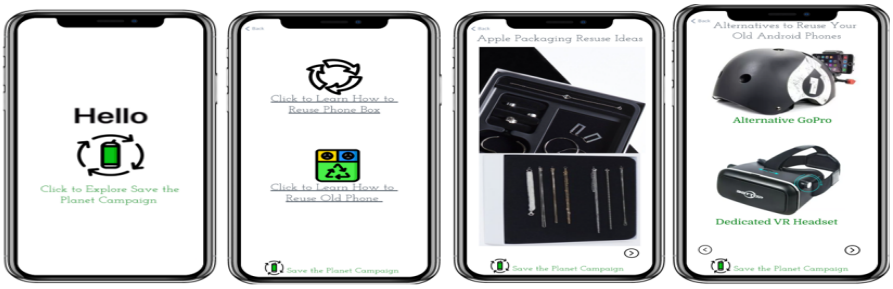


Fig. 6. Mobile phone setup prototype for Save the Planet Campaign Available Online [37]

Figure 6 shows some snapshots of the prototype, and the full version of the Save Planet Campaign prototype is available online [37]. Multinational technology companies like Apple and Samsung that contribute to the problems of e-waste and are leaders in the production and selling of different mobile devices and laptops can incorporate different campaigns similar to the Save the Planet Campaign in their electronic device setup to educate consumers or users of their products about sustainability and alternative reuse options for their electronic devices.

Next Action: Test the prototype (Save the Planet Campaign) and the three design mock-ups with end users to evaluate the effectiveness of supporting green UX among users.

4.4 Relevance Cycle 4: Testing prototype with end users

This section details the user experience testing of the Save the Planet Campaign prototype from section 4.3 with end users. The goal is to test the green UX based on user interaction with the prototype. Users were recruited by the researchers (authors) with different usage of ICT and technology needs. The volunteer users were asked the following questions before interacting with the Save the Planet Campaign prototype to gather information about each volunteer user and their habits:

- **Q1:** How many old smartphones do you have but currently do not use?
- **Q2:** What do you do with your old smartphones after you purchase new ones?
- **Q3:** What do you do with your phone package boxes after you purchase a new phone?

The responses from users for each of these questions are detailed in Table 6, with information on each participating volunteer user.

Table 6. Volunteered users information and answers before interacting with the Save the Planet Campaign prototype

Users	Age	Q1	Q2	Q3
User 1	52	2	I sometimes give it to my kids	I throw the package box to trash bin

User 2	50	3	Keep at home for memories	Keep them in the drawer
User 3	47	1	Throw them away since the batteries are usually bad	I save the boxes in my storage room
User 4	45	1	Dispose it off	Usually put in the trash
User 5	33	2	Left it in my drawer at our house	I leave it in my store
User 6	25	2	Put it in my garage storage	Trash them
User 7	23	1	Throw it away in the electronic bin	Put the package boxes in the recycle bin after unboxing new phones
User 8	21	0	Trade in when a new phone is released	Put them in the trash
User 9	20	0	Exchange for a better phone with an improved camera at the phone shop	Give my kid brother to play with it

The user test was conducted on 1st November 2023 with 9 volunteer users to interact and test the mobile phone setup for the Save the Planet Campaign. The volunteer users were provided with the prototype. All 9 volunteer users explored the prototype for an average of 15 minutes each. The following were verbatim comments from some of the volunteer users that summarize their opinions after exploring the mobile phone setup prototype for the Save the Planet Campaign:

- *Reflecting after this test I realized I can do more to reduce my electronic waste generation which can impact the environment*
- *I learned a lot of interesting ways in how my mobile phone package box and old smartphones can be put to good alternative use. My old smartphone can be a permanent GPS in my car, and in the future, I can use my old phone as a security camera or baby monitor when I have kids.*
- *There are alternative ways to reuse old smartphones. For example, reuse it as artwork, a security camera, and a virtual remote control.*
- *Now I know how to reuse my old mobile phone for useful purposes, not just to throw them away. Also, I got a new idea to use my phone box package as a jewelry holder.*
- *I will transform my phone boxes into jewelry and stationery boxes and use them as a key storage holder for my children and myself.*

The overall comments from the volunteer users indicated that they had learned something new through their experience of interacting with the prototype. However, it is one thing to learn something new and another to put it into daily practice. To test how this experience has influenced the user behavior, we deliberately waited three months to see what has changed in the user behavior.

The second phase of the user test was to identify changes in the user behavior towards alternative usage of old smartphones and phone boxes based on their sustainability awareness through the "Save the Planet Campaign." All the volunteer users were invited back after three months on 1st February 2024 to discuss changes in their usage of old smartphones and alternative reuse of mobile phone package boxes after unboxing new phones. The 9 volunteer users were asked the same questions as in the first phase. Their responses are detailed in Table 7. The responses from the volunteer users

(Table 7) show differences in user awareness and behavior (habits) when compared to responses in Table 6. All the volunteer users with at least one old smartphone have put it to alternative reuse (see responses in columns Q1 and Q2 in Tables 6 and 7).

Table 7. User test information and answers after prototype testing

Users	Age	Q1	Q2	Q3
User 1	52	1	I mount one of the old Android phones permanently on my car dashboard as a GPS	Phone boxes can be reused as a gift box.
User 2	50	1	My old mobile phone now serves as a baby monitor for my grandson and the other as a security camera.	As a plastic plant holder.
User 3	47	0	I have transformed my old phone into a bedside voice command alarm. I am using the wakeVoice application. I wonder why I did not think of this before now.	It can be used as a drawer divider to store small items.
User 4	45	0	I use the old smartphone as a second remote control in the house and also use it to stream videos to my LG TV.	Used as a stationary box on my kids' table.
User 5	33	1	The old smartphone is an awesome wireless mouse and keyboard that uses the unified remote app. I am working on making an artwork and asking my friends to donate their old smartphones.	Transformed to Jewelry holder and bracelet Storage.
User 6	25	0	Thanks to you (authors) for the ideas in the prototype. Now, I use my two old phones as security cameras for my room and storage using the WardenCam application.	Gift box and key holders on the shelve.
User 7	23	0	My old phone is now my ghost phone that hackers can't access. I will be using it for any of my bank transactions. I went through the links in the prototype and studied more to learn how to create untraceable phones.	Plastic flower vase for home and garden decoration.
User 8	21	0	You can use it in several ways, such as a security camera or remote control, or donate it to those in need.	Jewelry holder.
User 9	20	0	Use it as a cheap alternative to GoPro with phone mounts.	Stationary divider for pens, pencils, and a gift box.

Figure 7 shows an image of reusing an old smartphone as a permanent GPS in the car by volunteer User 1 (on the left), and User 5 shared a picture of reusing an iPhone package box as a jewelry box on the right. The results in this second phase indicate the potential for users to adopt greener habits through a sustainability-focused user experience, facilitated by ICT solutions. However, during our interaction with the volunteer users, we identified convenience as a barrier for transitioning to more sustainable habits as they are accustomed to certain routines in their daily activities.



Fig. 7. User 1 Shared an image of reusing an old smartphone as a permanent GPS in a car, and User 5 Shared an image of reusing a phone box as a jewelry box.

Some of the volunteer users expressed that their experiences during the first phase (test of the Save the Planet Campaign prototype) provided them with the awareness and knowledge to understand their impacts on sustainability, especially waste generation which prompted them to reconsider some of the daily habits. Therefore, ICT solution designers should take responsibility for integrating sustainability into their solutions to design sustainability-centric interactive software and services because they play a vital role in raising sustainability awareness and influencing user behavior toward sustainability.

5 Discussion and Threat to Validity

5.1 Discussion

Green user experience involves how well interactions between users of different ICT interactive software and services create user awareness about their impacts on sustainability. The GaLaPaGoS design patterns serve as a tool to support ICT designers in incorporating green and sustainability concerns into the design of ICT interactive software and services. User awareness about the current ecological problems has increased end users' demand for sustainability in ICT products and services [38]. The design of user experiences centered on green UX with sustainability can offer more value to users and companies. The GaLaPaGoS design patterns from this research support such design to aid more awareness about the ecological issues of our society and support design for change in user habits towards sustainability.

Several studies on persuasive technologies and patterns as reported in [25] [24] have explored self-centered values to influence user behavior. However, this approach can be counterproductive to sustainability efforts since self-enhancement persuasive technology often prioritizes personal benefits over fostering empathy-driven, long-term sustainable behaviors through technology [24]. In contrast, the GaLaPaGoS design patterns provide designers of interactive software and service a template for effectively engaging users to raise sustainability awareness and promote a sense of care for the environment. The findings from the developers and user test in sections 4.3 and 4.4

further illustrate that implementing the Galapagos design patterns can support design for green UX to foster internal motivation for sustainability among users for long-term sustainable behaviors.
Additionally, ICT interactive software and services for green UX covering user interactions and their awareness about their impacts on sustainability are less investigated. The challenge for software development practitioners, as expressed in the Participatory Pattern Workshop, is that most ICT designers and developers interested in sustainability lack concrete examples and tools that can demonstrate how sustainability is translated into design decisions in ICT solutions for green UX. Furthermore, to promote sustainability consciousness among users, interactive ICT software and services companies have the responsibility to incorporate means of gathering sustainability and green requirements for their ICT solutions to improve user sustainability awareness. The negative UX-based discovery approach is one method for gathering sustainability and green requirements[39]. However, there are fewer green UX patterns to support designers in incorporating those sustainability requirements in interactive software and service design. The GaLaPaGoS design patterns are created to address these design issues.

5.2 Threat to validity

- **Construct Validity:** To mitigate threats to construct validity, the research question was based on the results of the participatory pattern workshop with ICT practitioners from industry and academic publications related to the design of ICT interactive software and services for green UX.
- **Internal Validity:** The GaLaPaGoS design patterns were co-created by the authors and ICT practitioners in a participatory pattern workshop by analyzing problems of green UX from literature combined with industry challenges for green UX in ICT interactive software and services.
- **External Validity**: The GaLaPaGoS design patterns were validated by software developers and end users outside the participatory pattern workshop to ensure the generalizability of the results. The GaLaPaGoS design patterns show promising value in guiding the design of ICT interactive software and services for sustainability awareness among end users. However, Green UX requires more than the three patterns covered in this paper. We acknowledge a need for more testing of GaLaPaGoS design patterns by software development practitioners from different domains to collect data for improving and extending GaLaPaGoS design patterns. This will be done in future work to improve the generalization of results.

6 Conclusion

Green UX design is a new topic area, and there is no well-known best practice for ICT designers or software developers. Most developers and designers do not understand how and what it means to design, develop, and engineer ICT interactive software and services for green UX. These are the reasons for developing design patterns to capture best practices to guide software development practitioners during the design of ICT interactive software and services for green UX. Incorporating sustainability and green UX considerations in the design of ICT interactive software and services

will facilitate sustainability awareness among users and ignite their consciousness about their impacts while using different ICT solutions.
The three GaLaPaGoS design patterns, the design mock-ups created by software developers and tested by recruited users, demonstrate a real example of how designers can raise sustainability awareness among the end users at different stages of interactive ICT software and services. Based on the feedback from the developers, we observed that there is a need to create more patterns that guide and support designers of interactive software and services in presenting information within their design solution in a way that supports sustainability awareness and encourages reuse among users. One central and challenging problem is making the users of ICT interactive software and services aware of their negative actions toward the environment. This consists of exploring avenues to incorporate green UX and foster research towards making sustainability the core of ICT interactive software and services design. As part of plans for future work, we will improve the three GaLaPaGoS design patterns based on the feedback and organize a second participatory pattern workshop with software development practitioners to co-design and evaluate new patterns.

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